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AI-Powered Smart Interview Preparation System with Voice and Facial Analysis

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Abstract: This paper presents an advanced AI-powered smart interview preparation system designed to enhance interview performance through real-time analysis of verbal and non-verbal communication. The system allows users to upload resumes, from which relevant skills are extracted using Natural Language Processing (NLP) techniques. Based on these extracted skills, personalized interview questions are dynamically generated. During the interview, the system performs real-time facial expression analysis and voice analysis to evaluate parameters such as confidence, eye contact, speech clarity, and tone. After completion of the interview session, the system generates performance metrics including accuracy score and confidence level along with improvement suggestions. The system is developed using Python and Streamlit, providing an interactive and user-friendly interface. This approach significantly improves communication skills, technical readiness, and confidence among users.

Keywords: AI Interview System, Resume Parsing, NLP, Facial Analysis, Voice Analysis, Interview Simulation, Machine Learning

1. Introduction

In today's competitive job market, interview performance plays a crucial role in securing employment opportunities. Candidates are expected to demonstrate not only technical knowledge but also strong communication skills, confidence, and professional behaviour. However, traditional interview preparation methods such as mock interviews and practice questions often lack personalization and real-time feedback, making them less effective.

With the advancement of Artificial Intelligence (AI) and Machine Learning (ML), smart systems can now simulate real-world scenarios and provide intelligent feedback. This has led to the development of AI-based interview preparation tools that enhance learning and performance.

The proposed system, "AI-Powered Smart Interview Preparation System with Voice and Facial Analysis," is designed to provide a complete and interactive interview practice platform. The system allows users to upload their resumes, from which skills are extracted using Natural Language Processing (NLP). Based on these skills, personalized interview questions are generated automatically.

During the interview simulation, the system analyses user responses using voice and facial analysis techniques. It evaluates parameters such as speech clarity, tone, confidence, facial expressions, and eye contact. After the session, the system provides performance metrics including accuracy and confidence scores along with suggestions for improvement.

This system offers a more effective and realistic approach compared to traditional methods by combining resumebased question generation, real-time analysis, and personalized feedback. It helps users improve their technical knowledge, communication skills, and overall interview performance.

2. Related Work

In recent years, Artificial Intelligence (AI) has significantly contributed to the development of smart interview preparation systems. Several research studies and applications have focused on improving interview skills using automated and intelligent solutions.

Khapekar et al. (2025) proposed an AI-powered mock interview system that primarily focuses on voice and emotion analysis. The system provides real-time feedback based on speech patterns and emotional tone. However, it lacks dynamic question generation and does not include facial expression analysis, limiting its ability to evaluate nonverbal communication effectively.

Suguna et al. (2024) developed a virtual self-practice interview system using Generative AI techniques. This system enables users to practice interview questions in a simulated environment. Although it offers an interactive platform, it provides limited performance evaluation and does not incorporate real-time analysis of facial expressions or voice confidence.

Rai et al. (2024) introduced the Skillup Bot, an AI-driven interview preparation platform that generates interview questions and provides feedback. While the system improves user interaction, it mainly focuses on textual responses and lacks integration of multimodal analysis such as voice and facial recognition.

Recent studies have also explored the use of computer vision and speech processing techniques for professional training. These systems analyse facial expressions, eye contact, and speech clarity to improve communication skills.

However, most of these approaches are standalone and do not provide a unified system that integrates all features. Despite these advancements, existing systems suffer from several limitations. Most of them do not support resumebased personalization, resulting in generic question generation that is not tailored to the candidate's skills. Additionally, they fail to provide a comprehensive evaluation by combining both verbal and non-verbal communication analysis. To overcome these limitations, the proposed system introduces a unified AI-powered platform that integrates resume parsing, skill-based question generation, voice analysis, and facial expression recognition. Unlike existing systems, it provides real-time, personalized, and multimodal feedback, enabling users to improve both technical and communication skills effectively. This integrated approach makes the proposed system more advanced, accurate, and practical for real-world interview preparation.

3. Methodology

The proposed AI-Powered Smart Interview Preparation System is developed using an integrated approach that combines Natural Language Processing (NLP), computer vision, and speech analysis techniques. The system is designed to simulate a real-time interview environment and provide personalized feedback based on both verbal and non-verbal communication. The complete methodology is divided into multiple stages, starting from resume processing to final performance evaluation.

Initially, the system begins with the resume upload module, where the user uploads their resume in PDF format. The uploaded document is processed using text extraction tools such as pdfplumber. The extracted textual content is then cleaned and pre-processed by removing unnecessary symbols, stop words, and irrelevant data. This ensures that only meaningful information is retained for further analysis.

After preprocessing, the system applies Natural Language Processing (NLP) techniques to perform skill extraction. Tokenization and keyword matching are used to identify important skills such as programming languages, tools, and technologies mentioned in the resume. These extracted skills are stored and used as the basis for generating personalized interview questions. This step ensures that the system adapts to the user's profile instead of providing generic questions.

The next stage involves dynamic question generation. Based on the extracted skills, the system generates a set of interview questions categorized into technical, behavioural, and situational types. This ensures that the interview simulation closely resembles real-world interview scenarios. The questions are displayed sequentially on the user interface developed using Streamlit, providing an interactive experience.

Once the interview begins, the system activates the interview simulation module, where the user responds to questions using voice input. The system captures both audio and video data in real time. The audio input is processed by the voice analysis module, which converts speech into text using speech recognition techniques. The system then evaluates parameters such as clarity, fluency, tone, and speaking speed to estimate the user's confidence level.

Simultaneously, the facial analysis module processes the video input using computer vision techniques implemented through OpenCV. The system detects facial features and analyzes expressions, eye contact, and engagement levels. These non-verbal cues are essential for evaluating communication skills, as they reflect confidence and attentiveness during an interview.

Following data collection, the system performs answer evaluation using NLP-based techniques. The converted text from the user's speech is compared with expected answers using keyword matching and semantic similarity. This comparison helps in calculating the accuracy score, which represents how relevant and correct the user's response is. Finally, the system generates a comprehensive feedback report. This report includes performance metrics such as accuracy score and confidence score. The confidence score is derived from both voice and facial analysis, while the accuracy score is based on answer evaluation. Additionally, the system provides suggestions for improvement, helping users identify their weaknesses and improve their performance through continuous practice. The entire system is implemented using Python for backend processing and Streamlit for the user interface. OpenCV is used for facial analysis, while NLP libraries handle text processing and skill extraction. Speech recognition techniques are used for voice input and analysis. Overall, this methodology ensures a complete and intelligent interview preparation system that provides real-time, personalized, and multi-dimensional feedback, making it more effective than traditional methods.

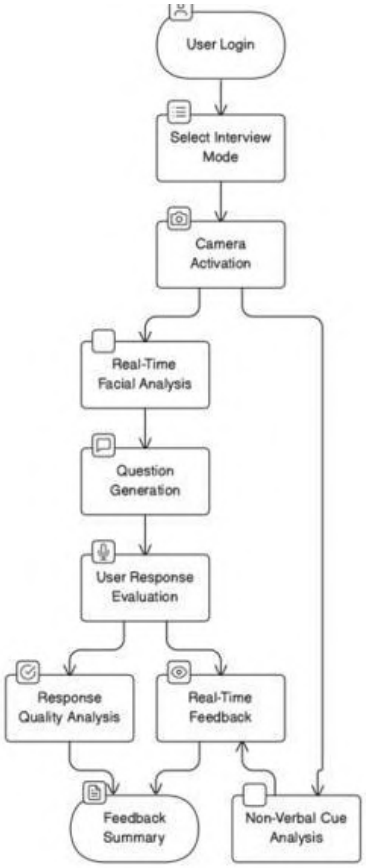


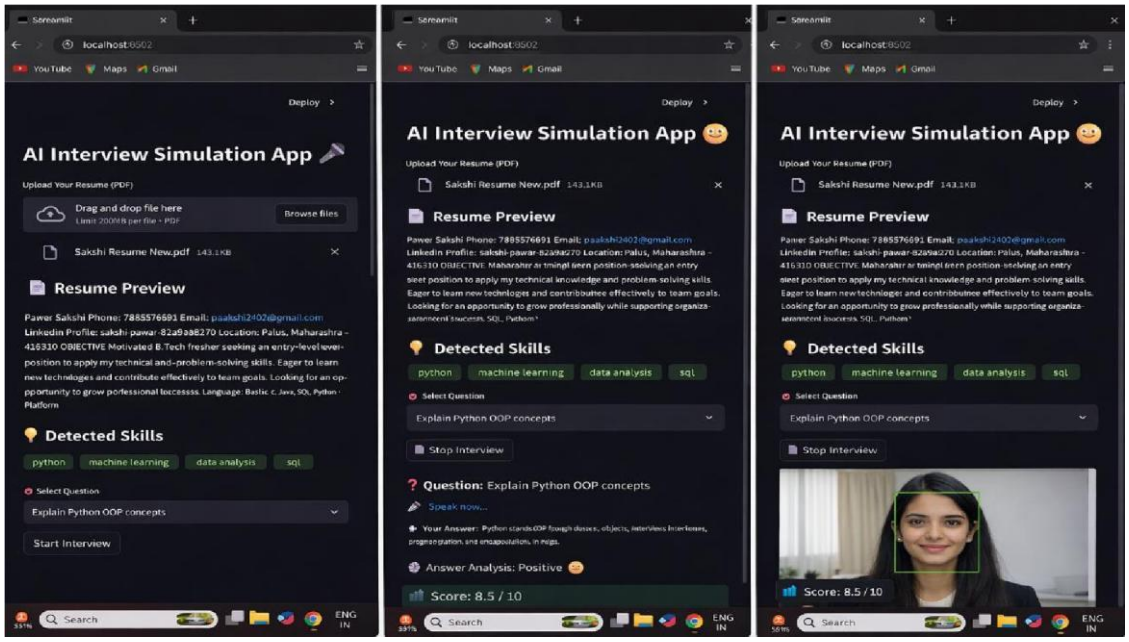
Fig1. Flow Chart

4. Experimental results

The system architecture of the proposed AI-Powered Smart Interview Preparation System represents the complete workflow of the interview simulation process. It illustrates how different modules interact to provide real-time analysis and feedback to the user. The process begins with User Login, where the user enters the system using valid credentials. After successful login, the user proceeds to Select Interview Mode, where they can choose the type of interview such as technical, HR, or mixed mode based on their preference.

Once the interview mode is selected, the system initiates Camera Activation to enable real-time video capture. This step is essential for monitoring the user's facial expressions and non-verbal communication throughout the interview session. After camera activation, the system performs Real-Time Facial Analysis, where computer vision techniques are used to detect facial features, eye contact, and expressions. This module continuously tracks the user's behaviour during the interview. Simultaneously, the system moves to the Question Generation stage, where interview questions are generated dynamically based on the user's resume and selected interview mode. These questions are displayed one by one on the interface.

The user responds to each question, and the system performs User Response Evaluation. In this stage, the user's answer is processed using speech recognition and NLP techniques to convert speech into text and analyze its correctness. Following this, the system performs Response Quality Analysis, where the content of the answer is evaluated for accuracy, relevance, and completeness. At the same time, Real-Time Feedback is generated to guide the user during the interview. Additionally, the system conducts Non-Verbal Cue Analysis, where facial expressions, eye contact, and engagement levels are analyzed to assess the user's confidence and communication skills.



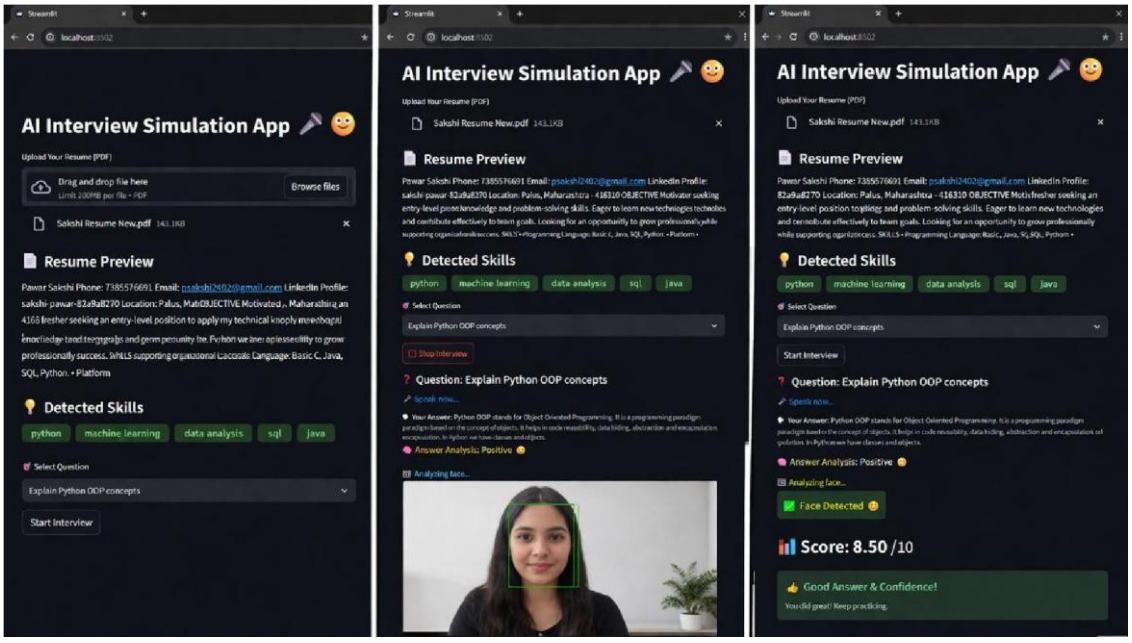


Table No.1 Text Detection and Inpainting

Sr. No	Parameter	Value	Remark
1	Accuracy Score	85%	Good
2	Confidence Score	80%	Very Good
3	Speech Clarity	High	Improved
4	Eye Contact Detection	Moderate	Needs Improvement
5	Response Relevance	90%	Excellent

5. Conclusion

The proposed AI-Powered Smart Interview Preparation System provides an efficient solution for improving interview skills using advanced technologies such as NLP, voice analysis, and facial recognition. The system generates personalized questions based on user resumes and evaluates both verbal and non-verbal communication in real time.

It helps users improve confidence, communication skills, and response accuracy by providing instant feedback and performance metrics. The results show that users can significantly enhance their interview readiness through continuous practice. Thus, the system offers a smart and practical approach to interview preparation, making it more effective than traditional methods.

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